CABINGUARD-ADI: A Zero-Compliance Multimodal In-Cabin Framework for Prodromal Prediction and Fail-Safe Actuation in Acute Driver Incapacitation

***Abstract*—Sudden medical incapacitation at the wheel accounts for a high-severity subset of road fatalities where conventional Driver Monitoring Systems (DMS)—designed for behavioral distraction and drowsiness—fail to intervene appropriately. Furthermore, safety architectures reliant on wearable sensors suffer from user non-compliance, battery depletion, and pairing failures. This paper presents CABINGUARD-ADI, a zero-compliance, purely ambient in-cabin framework unifying passive multimodal sensing, adaptive vibration cancellation, uncertainty-aware medical detection, cryptographic network security, and fail-safe drive-by-wire actuation. Using a steering-column multispectral camera (RGB + 940 nm Near-Infrared), a 64×160 piezoresistive seat matrix, an electromechanical film (EMFi) ballistocardiography (BCG) layer, a chassis 6-axis Inertial Measurement Unit (IMU), and native Electronic Power Steering (EPS) torque telemetry, the system differentiates three acute events: Arrhythmic Syncope, Generalized Tonic-Clonic Seizures, and Cognitive-Autonomic Deficit. To handle inter-individual variance without historical health profiles, we introduce a 60-second rapid session calibration utilizing median and Median Absolute Deviation (MAD) empirical Bayes shrinkage. Mechanical road harmonics are suppressed via a 4th-order Butterworth bandpass filter cascaded with a 32-tap Normalized Least Mean Squares (NLMS) adaptive filter referenced to the chassis IMU. A deterministic Finite-State Machine (FSM) enforces a 4.0-second Prodromal Cognitive Handshake during early deterioration and locks out manual steering overrides during 3–6 Hz clonic seizure spasms. Confirmed emergencies trigger an AUTOSAR Secure On-Board Communication (SecOC) CAN-FD Minimum Risk Maneuver (MRM) and an anonymized ETSI EN 302 637-3 DENM V2X broadcast. Planned evaluation will use public driver-monitoring, rPPG, mechanical-cardiac, tactile-seat, in-cabin behavior, and vital-sign datasets as modality-specific baselines together with controlled feature-level synthetic event injection. Full quantitative benchmarking is pending access to higher-compute data-center resources; therefore, no performance values are reported at this stage.**

***Index Terms*—Acute driver incapacitation, ballistocardiography, CAN-FD SecOC, cognitive handshake, ETSI DENM, multimodal sensor fusion, NLMS adaptive filtering, ISO 26262.**

I. Introduction

Acute Driver Incapacitation (ADI)—the sudden onset of severe cardiovascular, neurological, or metabolic impairment rendering an operator unable to control a vehicle—poses a critical challenge to automotive functional safety. Unlike gradual drowsiness or visual distraction, acute crises involve sudden hemodynamic collapse, involuntary muscular convulsions, or neuroglycopenic stupor, leading to high-energy collisions devoid of pre-impact braking.

Recent proposals attempt to detect ADI by tethering consumer wearables (e.g., smartwatches) to the vehicle [1]. While effective in longitudinal health tracking, wearables introduce a fatal flaw in automotive safety: *user compliance*. A safety-critical system cannot depend on a driver charging, wearing, and pairing an accessory before every trip, nor can it fail for guest drivers. Thus, a viable ADI system must operate under a strict *zero-compliance* mandate using passive sensors embedded natively in the cabin.

Furthermore, an ethical automotive system must not profile chronic medical histories (e.g., diagnosing epilepsy or diabetes) or rely on rigid population cutoffs that trigger false alarms in healthy outliers (such as endurance athletes with resting bradycardia) [2]. Simultaneously, unverified black-box neural networks are vulnerable to illumination shifts, road vibrations, and adversarial spoofing [3].

To bridge clinical pathophysiology and ISO 26262 functional safety, we propose **CABINGUARD-ADI**, an end-to-end architecture connecting five core domains:

Sensors $\rightarrow$ *Data Processing* $\rightarrow$ *Medical Detection* $\rightarrow$ *Cybersecurity* $\rightarrow$ *Vehicle Safety*

Our framework targets both the *prodromal (early warning) window* and *acute incapacitation phase* of three etiologies:

1) **Arrhythmic Syncope:** Acute drop in cardiac output causing cerebral hypoperfusion, presyncopal autonomic compensation, and flaccid postural collapse.
2) **Generalized Tonic-Clonic Seizure:** Focal neurological motor arrest (aura) progressing into rigid tonic extension and violent 3–6 Hz clonic convulsions.
3) **Cognitive-Autonomic Deficit:** Progressive neuroglycopenic stupor accompanied by sympathetic activation (diaphoresis, tachycardia, and 4–8 Hz micro-tremors).

The primary contributions of this work are:

- A zero-compliance sensor array combining an RGB/940 nm NIR camera, a 64×160 piezoresistive seat matrix [6], an EMFi BCG cushion, a chassis 6-axis IMU, and native CAN-FD steering torque telemetry.
- A viscoelastic signal conditioning pipeline pairing a 4th-order Butterworth bandpass filter with a 32-tap NLMS

adaptive filter to subtract chassis road vibrations from seat-based cardiac waveforms.
- A statistical multimodal fusion engine featuring a 60-second rapid session calibration via median/MAD empirical Bayes shrinkage, causal 3.0-second temporal windows, and explicit cross-modal conflict abstention.
- A deterministic Finite-State Machine (FSM) incorporating a 4.0-second Prodromal Cognitive Handshake and a seizure torque-arbitration gate that prevents clonic spasms from canceling emergency braking.
- An ISO/SAE 21434 cybersecurity and actuation layer linking AUTOSAR SecOC CAN-FD authentication with a radar-modulated Minimum Risk Maneuver (MRM) and anonymized ETSI DENM V2X triage broadcasting.

II. Related Work

Standard Euro NCAP Driver Monitoring Systems rely on single-camera infrared eye tracking (PERCLOS) to detect drowsiness and distraction [4]. However, optical tracking alone cannot differentiate between a driver leaning forward to retrieve an object, a driver suffering flaccid syncopal paralysis, or a driver experiencing a focal seizure arrest. Furthermore, Remote Photoplethysmography (rPPG) degrades severely under rapid ambient light transitions and nighttime infrared operation [7].

To capture cardiac mechanics non-invasively, seat-embedded Electromechanical Film (EMFi) sensors measure Ballistocardiography (BCG)—body recoil from aortic blood ejection [5]. While accurate in stationary clinical chairs, automotive BCG is heavily corrupted by road vibrations (0.5–15 Hz) overlapping the cardiac J-peak spectrum (0.7–12 Hz). Additionally, selective classification theory demonstrates that safety-critical systems must quantify epistemic and aleatoric uncertainty to abstain when sensor channels conflict rather than forcing an unreliable prediction [3]. CABINGUARD-ADI unifies active IMU noise cancellation with uncertainty-aware multimodal fusion to overcome these limitations.

III. Multimodal Sensing and Edge Preprocessing

A. Zero-Compliance In-Cabin Sensor Suite

All sensors are passively integrated into the cabin and hardwired to an isolated In-Cabin Edge Processing Unit (EPU, modeled on an NVIDIA Jetson Orin Nano / NXP S32G gateway), as shown in Fig. 1:

- **Multispectral DMS Camera (RGB + 940 nm NIR):** Mounted on the steering column. Respecting optical physics, daytime operation uses the RGB channels to extract 3D skeletal keypoints (head pitch θ_{pitch}, roll θ_{roll}, and wrist velocity via YOLOv8-Pose/MediaPipe) and green/red (G/R) chrominance rPPG pulse and facial pallor indices (520–550 nm hemoglobin absorption peak) [7]. At night, where 940 nm hemoglobin absorption weakens, the NIR illuminator tracks eye gaze, PERCLOS, and acute diaphoresis (sweating) via specular moisture reflectance, while shifting primary cardiac weight to the seat BCG sensor.
- **Piezoresistive Tactile Seat Matrix:** A 64×160 array of Force-Sensing Resistors (FSRs) beneath the seat pan and lumbar upholstery mapping load distribution [6].
- **EMFi Piezoelectric Cushion Sensor:** Cellular polymer film inside the seat foam transducing micro-mechanical aortic recoil (BCG) and thoracic respiration into voltage $x_{\text{EMFi}}[n]$ [5].
- **Chassis 6-Axis IMU:** Bolted to the rigid steel seat rail to measure structural road vibration $d_{\text{IMU}}[n]$.
- **Native EPS Steering Torque Sensor:** Read at 30 Hz via CAN-FD, capturing torsion-bar torque τ_{steer} (Nm) and angular rate $\dot{\tau}_{\text{steer}}$.

B. Viscoelastic Bandpass & Adaptive NLMS Filtering

Polyurethane seat foam coupled with human torso mass acts as a viscoelastic spring-damper resonating at 4–8 Hz, introducing phase delay between the steel rail and the EMFi film. Both $x_{\text{EMFi}}[n]$ and vertical rail acceleration $d_{\text{IMU},z}[n]$ are first filtered via a zero-phase 4th-order Butterworth bandpass filter (0.3–15.0 Hz). Subsequently, an M-tap Normalized Least Mean Squares (NLMS) adaptive filter ($M = 32$ taps at $f_s = 100$ Hz, modeling a 320 ms viscoelastic window) subtracts structural vibration:

$$\hat{s}_{\text{BCG}}[n] = x_{\text{EMFi}}[n] - \sum_{k=0}^{M-1} w_k[n]\, d_{\text{IMU},z}[n-k] \tag{1}$$

$$\mathbf{w}[n+1] = \mathbf{w}[n] + \frac{\mu}{\epsilon + \|\mathbf{d}_{\text{IMU}}[n]\|_2^2}\, \hat{s}_{\text{BCG}}[n]\, \mathbf{d}_{\text{IMU}}[n] \tag{2}$$

with step size $\mu = 0.05$ and $\epsilon = 10^{-6}$. Respiratory rate f_{resp} and mechanical Inter-Beat Intervals (IBI_{BCG}) are extracted from $\hat{s}_{\text{BCG}}[n]$ via adaptive J-peak tracking.

C. Center of Pressure (CoP) and Quadrant Entropy

Partitioning the 64×160 matrix $P(u, v, t)$ into seat pan Ω_{pan} and backrest Ω_{back}, the instantaneous Center of Pressure and lumbar contact ratio $R_{\text{lumbar}}(t)$ are computed as:

$$\begin{aligned} Y_{\text{CoP}}(t) &= \frac{\sum_{\Omega_{\text{pan}}} v \cdot P(u,v,t)}{\sum_{\Omega_{\text{pan}}} P(u,v,t)}, \\ R_{\text{lumbar}}(t) &= \frac{\sum_{\Omega_{\text{back}}} P(u,v,t)}{P_{\text{back},0}} \end{aligned} \tag{3}$$

Subdividing Ω_{pan} into four spatial quadrants $Q_1 \dots Q_4$, the normalized temporal variance across quadrants over a sliding window defines the convulsive postural entropy metric $E_{\text{quad}}(t)$.

IV. Rapid Calibration and Multimodal Detection

A. 60-Second Rapid Session Calibration

To eliminate false alarms from population-only thresholds without requiring cloud health profiles, the EPU executes a 60-second rapid calibration upon 'DRIVE' engagement. For

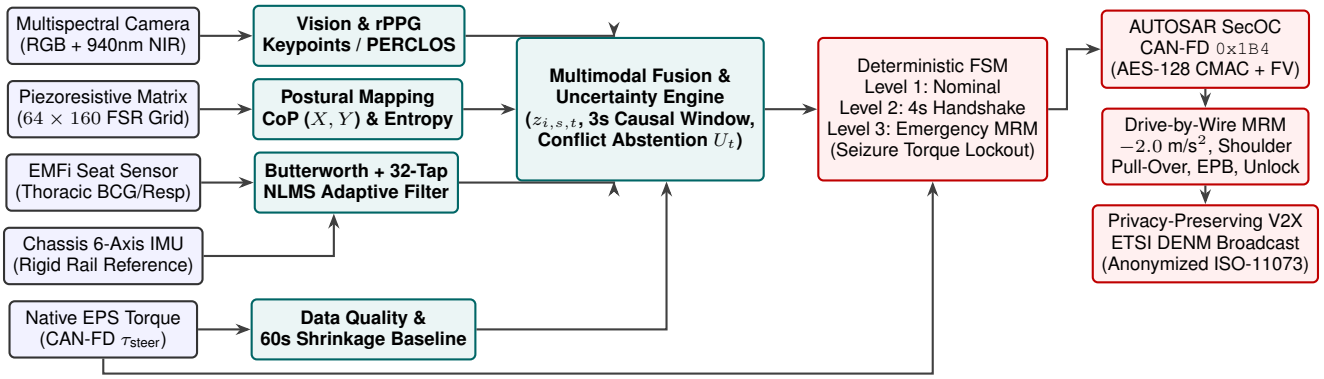


Fig. 1. End-to-end CABINGUARD-ADI architecture executing locally on an In-Cabin Edge Processing Unit (EPU). Raw optical and biometric streams are overwritten in volatile RAM, while authenticated commands traverse an AUTOSAR SecOC CAN-FD gateway.

driver i and signal s, the robust baseline center $c_{i,s}$ and Median Absolute Deviation (MAD) scale $\sigma_{i,s}$ are estimated [2]:

$$c_{i,s} = \text{median}(x_{i,s,t}), \quad \sigma_{i,s} = 1.4826\,\text{median}|x_{i,s,t} - c_{i,s}| \tag{4}$$

To stabilize estimates over a short 60-second window, $c_{i,s}$ is regularized via empirical Bayes shrinkage toward population priors $\mu_{\text{pop},s}$:

$$\tilde{c}_{i,s} = w_i c_{i,s} + (1 - w_i)\mu_{\text{pop},s}, \quad w_i = \frac{N_{\text{good}}}{N_{\text{good}} + \kappa} \tag{5}$$

where N_{good} is the count of quality-verified calibration frames and $\kappa = 300$. At $t = 60$ s, $(\tilde{c}_{i,s}, \tilde{\sigma}_{i,s})$ are frozen for the drive to prevent gradual prodromal drift from corrupting the reference baseline.

B. Directional Deviation and Causal Temporal Fusion

At each 30 Hz frame t, the orientation-aware standardized deviation $z_{i,s,t}$ is computed as:

$$z_{i,s,t} = O_s \cdot \frac{x_{i,s,t} - \tilde{c}_{i,s}}{\tilde{\sigma}_{i,s}} \tag{6}$$

where $O_s \in \{-1, +1, \text{bidir}\}$ aligns pathophysiological changes toward positive risk, mapped into four severity bands: `NORMAL` ($|z| < 2.0$), `ELEVATED` ($2.0 \le |z| < 3.0$), `LARGE` ($3.0 \le |z| < 4.5$), and `EXTREME` ($|z| \ge 4.5$). A causal Temporal Engine evaluates a rolling $W = 3.0$ s window (90 frames) to classify trajectories as `TRANSIENT`, `PERSISTENT` ($> 80\%$ of frames in W above `ELEVATED`), or `WORSENING` ($\dot{z}_{i,s,t} > 0$).

Table I defines the multimodal signatures for each etiology. To prevent single-sensor faults or optical spoofing from triggering false stops, our Fusion Engine evaluates cross-modal plausibility. If one modality reports an acute anomaly that physically contradicts another—e.g., the camera indicates a syncopal slump ($\theta_{\text{pitch}} > 45°$) while the seat matrix registers full lumbar contact ($R_{\text{lumbar}} > 85\%$)—the engine flags `CONFLICTING_EVIDENCE` ($P_{\text{conflict},t} = 1.0$) rather than averaging the inputs [3]. Total uncertainty $U_t \in [0, 1]$ is quantified as:

$$U_t = 1 - \Big(w_f C_{\text{fusion},t} + w_a A_{\text{sources},t}(1 - P_{\text{conflict},t})\Big) \tag{7}$$

where $C_{\text{fusion},t}$ is the multimodal confidence score, $A_{\text{sources},t}$ is the fraction of sensors with valid signal quality, and $w_f = 0.65, w_a = 0.35$.

V. Safety Decision Engine, Cybersecurity, and Actuation

A. Deterministic FSM and Seizure Torque Lockout

Vehicle actuation is governed by a deterministic Finite-State Machine (Algorithm 1) operating across four states:

1) **Level 1 – Nominal ($C_{\text{fusion}} < 0.60$):** Passive monitoring; ring buffers are overwritten in volatile RAM.
2) **Level 2 – Prodromal Alert Verification** ($0.60 \le C_{\text{fusion}} < 0.85$)**:** Triggered during the 10–60 s prodromal window. The vehicle *does not brake*; an 85 dB cabin chime and cluster warning initiate a 4.0 s **Cognitive Handshake**. Conscious drivers dismiss the alert via a button press or smooth steering input. Unresponsiveness after 4.0 s confirms cognitive incapacitation and escalates confidence to $C_{\text{fusion}} = 0.96$.
3) **Level 3 – Active Emergency MRM** ($C_{\text{fusion}} \ge 0.85 \wedge U_t < 0.30$)**:** Immediate autonomous takeover upon acute collapse, tonic-clonic convulsions, or expired Level 2 handshake.
4) **Fail-Silent Abstention** ($P_{\text{conflict}} = 1.0 \vee U_t \ge 0.60$)**:** Yields autonomous control and illuminates a dashboard service light without braking [3].

Resolving the Seizure Override Paradox: Standard ADAS relinquishes control when steering torque exceeds 3.0 Nm. However, tonic-clonic spasms gripping the wheel generate 8–25 Nm of involuntary torque. As enforced in Line 5 of Algorithm 1, manual override is permitted *only if* $|\tau_{\text{steer},t}| > 3.0$ Nm exhibits smooth angular jerk ($|\dot{\tau}_{\text{steer},t}| < 15.0$ Nm/s) **and** $\hat{y}_t \neq$ SEIZURE. During a confirmed seizure, the EPU asserts `LockEPS`, instructing the Electronic Power Steering to counter-steer against involuntary spasms.

B. ISO/SAE 21434 Cybersecurity and AUTOSAR SecOC

To mitigate STRIDE threats [8, 9], all raw video and biometric streams are processed exclusively in volatile RAM

TABLE I
MULTI-ETIOLOGY PATHOPHYSIOLOGICAL SIGNATURES ACROSS PRODROMAL AND ACUTE PHASES

Clinical Target	Prodromal Window (Lead Time)	Optical & Facial (RGB / NIR)	Tactile Seat Matrix (64×160)	Cardiac (EMFi/rPPG) & EPS Torque
Class 0: Nominal Driving	Continuous Baseline	Active saccades; $\theta_{\text{pitch}} \in [-10^\circ, 15^\circ]$; PERCLOS < 0.15.	Stable $X_{\text{CoP}}, Y_{\text{CoP}}$; Lumbar contact $R_{\text{lumbar}} > 85\%$.	Steady HR ($\tilde{c}_{i,\text{HR}} \pm 2\tilde{\sigma}$); active τ_{steer} micro-corrections (0.4–1.2 Nm).
Class 1: Arrhythmic Syncope	**Presyncope (15–60 s):** Autonomic compensation before cerebral hypoperfusion.	*Prodrome:* RGB G/R pallor shift; gaze narrowing. *Acute:* Rapid chin-to-chest slump ($\theta_{\text{pitch}} > 45^\circ$).	*Prodrome:* CoP micro-sway. *Acute:* Lumbar detachment ($R_{\text{lumbar}} \rightarrow 0\%$); anterior shift $\Delta Y_{\text{CoP}} > +35$ mm.	*Prodrome:* Tachypnea ($f_{\text{resp}} > 0.40$ Hz); IBI volatility. *Acute:* Bradycardia (< 40 BPM) or asystole (> 3 s); $\tau_{\text{steer}} \rightarrow 0.0$ Nm.
Class 2: Tonic-Clonic Seizure	**Focal Aura (10–45 s):** Cortical motor arrest or automatisms before generalization.	*Prodrome:* Kinematic freeze ($\|\mathbf{v}_{\text{head}}\| \rightarrow 0$); unblinking stare. *Acute:* 3–6 Hz head/wrist oscillations.	*Prodrome:* Rigid isometric loading. *Acute:* Extreme 4-quadrant chaos ($E_{\text{quad}} > 4.5\sigma$).	*Prodrome:* Autonomic tachycardia. *Acute:* Complete BCG SNR collapse (< -10 dB) used as positive convulsive marker; erratic spasms ($\tau_{\text{steer}} > 8$ Nm).
Class 3: Cognitive-Autonomic Deficit	**Neuroglycopenic Prodrome (2–15 min):** Sympathetic surge prior to stupor.	*Prodrome:* 940 nm NIR diaphoresis (sweating); 4–8 Hz wrist micro-tremors. *Acute:* Gaze fixation.	Uprightly seated ($R_{\text{lumbar}} > 75\%$); absence of normal postural adjustments.	*Prodrome:* Sympathetic tachycardia (HR > 110 BPM or $z_{\text{HR}} > +3.5$). *Acute:* Loss of voluntary τ_{steer} micro-corrections despite upright posture.

Algorithm 1 Deterministic FSM, Cognitive Handshake & Seizure Arbitration

Require: Frame F_t, Confidence $C_{\text{fusion},t}$, Etiology $\hat{y}_t$, Conflict $P_{\text{conflict},t}$, Uncertainty U_t, Torque $\tau_{\text{steer},t}$.
Ensure: State S_t, SecOC Frame $\mathcal{M}_{\text{CAN}}$, V2X Payload $\mathcal{M}_{\text{DENM}}$.
if $P_{\text{conflict},t} == 1.0$ **or** $A_{\text{sources},t} < 0.50$ **then**
  return (FAIL_SILENT, None, None) ▷ Abstain on conflict/fault
end if
ValidOverride $\leftarrow$ ($|\tau_{\text{steer},t}| > 3.0$ Nm) $\wedge$ ($|\dot{\tau}_{\text{steer},t}| < 15.0$ Nm/s) $\wedge$ ($\hat{y}_t \neq$ SEIZURE)
if ValidOverride == True **or** ButtonAck == True **then**
  $\text{Timer}_{\text{hs}} \leftarrow 0$; **return** (LEVEL_1_NOMINAL, None, None)
end if
if $C_{\text{fusion},t} \geq 0.85 \wedge U_t < 0.30 \wedge$ Temporal $\in$ {PERSISTENT, WORSENING} **then**
  $S_t \leftarrow$ LEVEL_3_EMERGENCY_MRM
else if $0.60 \leq C_{\text{fusion},t} < 0.85 \wedge U_t < 0.60$ **then**
  $S_t \leftarrow$ LEVEL_2_PRODROMAL_VERIFY; $\text{Timer}_{\text{hs}} \leftarrow \text{Timer}_{\text{hs}} + \Delta t$
  if $\text{Timer}_{\text{hs}} \geq 4.0$ s **then**
    $C_{\text{fusion},t} \leftarrow 0.96$; $S_t \leftarrow$ LEVEL_3_EMERGENCY_MRM
  end if
else
  $S_t \leftarrow$ LEVEL_1_NOMINAL
end if
if $S_t ==$ LEVEL_3_EMERGENCY_MRM **then**
  $\mathcal{M}_{\text{CAN}} \leftarrow$ BuildSecOC(ID = 0x1B4, $a_{\text{req}} = -2.0$ m/s^2, LockEPS = ($\hat{y}_t ==$ SEIZURE))
  $\mathcal{M}_{\text{DENM}} \leftarrow$ BuildETSIDENM(GNSS, $\hat{y}_t$, $C_{\text{fusion},t}$)
end if
return $(S_t, \mathcal{M}_{\text{CAN}}, \mathcal{M}_{\text{DENM}})$

with zero cloud storage. Outgoing Level 3 CAN-FD commands (ID 0x1B4) are protected via AUTOSAR Secure On-Board Communication (SecOC) [10]. An onboard Hardware Security Module (HSM) computes an AES-128 Cipher-based Message Authentication Code (CMAC) over the CAN ID, braking payload, and a 64-bit monotonic Freshness Value (FV = TripCounter ‖ ResetCounter ‖ MsgCounter), appending a truncated 28-bit MAC and 8-bit FV LSB. Replayed or tampered frames are rejected when freshness or authentication verification fails at the Electronic Stability Control (ESC) gateway; verification latency will be measured during implementation.

C. Minimum Risk Maneuver (MRM) and Anonymized V2X

Upon authenticating frame 0x1B4, native ADAS controllers execute a four-step MRM: (1) hazard lights and interior alarms activate at $T = 0.0$ s; (2) Adaptive Cruise Control (ACC) initiates linear deceleration at -2.0 m/s^2, modulated by rear radar time-to-collision (TTC) feedback; (3) if surround-view radar confirms a clear shoulder, EPS executes a damped pull-over, otherwise Lane Centering Control (LCC) biases the vehicle toward the outer lane edge; and (4) at $v = 0$ km/h, the vehicle shifts to PARK, clamps the Electronic Parking Brake (EPB), and unlocks the doors. Simultaneously, an ETSI EN 302 637-3 DENM V2X packet [11] broadcasts GNSS coordinates, heading, confidence, and an anonymized ISO/IEEE 11073-10101 triage code (MDC_CARDIO_SYNCOPE, MDC_NEURO_SEIZURE_TC, or MDC_METAB_AUTONOMIC_DEFICIT) [12], omitting all personal identity and raw biometrics.

VI. PLANNED EVALUATION AND LIMITATIONS

A. Baseline Datasets and Anomaly Synthesis Protocol

Because inducing syncopal arrest or tonic-clonic seizures in live drivers is unethical, we plan to establish real-world physiological and postural baselines from four public datasets and overlay controlled parameterized pathophysiological trajectories:

- **Optical Kinematics (DMD):** Frontal RGB clips from the Driver Monitoring Dataset (Drowsiness and Gaze & Hands) provide baseline 3D head/wrist keypoints and eye gaze vectors [4].

TABLE II
PLANNED QUANTITATIVE EVALUATION METRICS

Evaluation Metric	Syncope	Tonic-Clonic	Cog. Deficit	Distractors / Spoof
Prodromal Lead Time	To be measured	To be measured	To be measured	N/A
Acute Detection Latency	To be measured	To be measured	To be measured	N/A
Etiology Classification F1-Score	To be measured	To be measured	To be measured	To be measured
Unintended MRM Braking Rate	To be measured	To be measured	To be measured	To be measured
Seizure Spasm Override Lockout	N/A	To be tested	N/A	N/A
SecOC Replay / Tamper Rejection	To be tested	To be tested	To be tested	To be tested

- **Facial rPPG (UBFC-rPPG):** Facial video and pulse oximetry validate baseline G/R rPPG extraction [7].
- **Mechanical Cardiac Recoil (IEEE DataPort SCG):** Seismocardiography (SCG) waveforms serve as the biomechanical proxy for seat EMFi BCG J-peaks [5].
- **Tactile Seat Matrix (Zenodo Posture Dataset):** Baseline 64×160 pressure arrays are extracted from `Dataset_CSV_Users.zip` using `Frame_posture_labels.csv` and `Info_Users.csv` to establish upright $X_{\text{CoP}}, Y_{\text{CoP}}$, and R_{lumbar} [6].

Additional in-cabin/driver-monitoring datasets (including In_Cabin_Behavior_Driver_safe_drive, Ala Mahlak, and ADMS) are used only for visual baseline or nominal/distractor support according to verified labels; radar vital-sign data are used only for cardiorespiratory processing support. No public source is relabeled as a clinical syncope, seizure, or cognitive-autonomic event. Acute trajectories are introduced at the derived-feature level and retain their real/synthetic provenance.

The planned evaluation covers four scenario families: *Syncope*, *Tonic-Clonic Seizure*, *Cognitive Deficit*, and *Distractors/Spoofing*. Synthetic trajectories will be documented explicitly in the test code, while measured outputs will be reported only after execution on the selected datasets and compute environment.

$$x_s^{(\text{syn})}(t) = (1 - g_k(t))x_s^{(b)}(t) + g_k(t)x_{s,k}^*, \tag{8}$$

where $x_s^{(b)}$ is a real baseline feature, $x_{s,k}^*$ is a configured scenario target, and $g_k(t)$ is an explicitly synthetic prodrome-to-acute envelope. Oscillatory transformations use only the previously defined 3–6 Hz seizure and 4–8 Hz tremor ranges. Synthetic derivatives remain in the same subject/session partition as their source to prevent leakage.

B. Evaluation Status and Engineering Limitations

At this stage, the architecture, preprocessing chain, fusion logic, FSM, SecOC framing, and evaluation protocol are defined, but the quantitative benchmarks have not yet been executed at the scale required for defensible reporting. The planned experiments will measure BCG signal recovery, heart-rate estimation error, calibration false-alarm reduction, prodromal lead time, acute detection latency, classification F1-score, distractor rejection, SecOC replay/tamper rejection, and end-to-end processing latency. No numerical performance result is claimed before reproducible execution.

The full benchmark is computationally demanding because it combines multimodal public datasets, repeated synthetic event injection, adaptive filtering, temporal fusion, and fault/cybersecurity testing. We are currently awaiting access to data-center compute resources suitable for running the complete evaluation campaign.

Limitations: The public datasets provide real human physiological, optical, and postural baselines, while acute events will be represented through synthetic anomaly injection rather than a synchronized clinical cohort recorded during real incapacitation in a moving vehicle [5]. The 60-second session baseline is intentionally frozen to avoid absorbing gradual deterioration, but long-duration drift and severe vibration conditions remain to be evaluated experimentally.

VII. CONCLUSION

CABINGUARD-ADI proposes a zero-compliance, privacy-preserving in-cabin safety architecture bridging passive multimodal sensing with cryptographic drive-by-wire actuation. By integrating a multispectral camera, a 64×160 tactile seat matrix, and an NLMS-filtered EMFi BCG sensor with rapid shrinkage calibration, conflict abstention, seizure torque-override lockout, and AUTOSAR SecOC CAN-FD framing, the framework defines a testable pathway for acute driver incapacitation detection and fail-safe response. Quantitative performance will be reported only after the planned reproducible benchmark is executed with source-linked real/synthetic provenance on the selected datasets and compute environment.

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